

Key

There are 150 points in the exam, across 12 numbered problems, some with multiple parts. The total number of points for each problem is indicated.

Each part of a problem is worth the same amount.

There is also a bonus problem on the last page of the exam.

1. (24 points) Give an example of each of the following, or explain why it is not possible.

a) A matrix A and two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ so that these vectors are eigenvectors of A but $\mathbf{v}_1 + \mathbf{v}_2$ is not an eigenvector of A .

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad A \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{but} \quad A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

b) Two square matrices that are column equivalent but not similar.

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

c) An operator A on $\mathbb{R}^3$ and a vector $\mathbf{v}$ so that $\mathbf{v}$ is in both the column space and the nullspace of A .

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

d) For some n , two orthogonal $n \times n$ matrices that are not similar to each other.

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

since they do not have equal determinants.

e) An $n \geq 1$ and a subspace V of $\mathbb{R}^n$ such that there does not exist a linear transformation with codomain $\mathbb{R}^n$ and range V .

Not possible. V has a finite basis, B . Set T to be a linear transformation represented by a matrix with $b_i \in B$ as its columns.

Then the range of A is V . (If V is the trivial subspace, any zero matrix with codomain $\mathbb{R}^n$ suffices.)

f) A 3×3 matrix A that has no square root. That is, there exists no B such that $B^2 = A$.

For instance ~~$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$~~ $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$,

which is nilpotent of rank 2.

g) A matrix that has 8 as an eigenvalue with algebraic multiplicity 5 and geometric multiplicity 3.

e.g. $\begin{pmatrix} \boxed{8} & 0 & 0 & 0 & 0 \\ 0 & \boxed{8} & 1 & 0 & 0 \\ 0 & 0 & \boxed{8} & 0 & 0 \\ 0 & 0 & 0 & \boxed{8} & 1 \\ 0 & 0 & 0 & 0 & \boxed{8} \end{pmatrix}$

h) A matrix with nullspace of dimension 2 and a left nullspace of dimension 3.

If $n \times m$, we have $\text{rank } A = m - 2$
and $\text{rank } A^T = n - 3$

Since these are equal, $m - n = 1$
So for instance, $\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$

2. (6 points) Give a precise definition for each of the following linear algebra terms.

nullspace:

The nullspace of a linear transformation $L: V \rightarrow W$ is the subset of V sent to the zero vector in W by L .

eigenvector and eigenvalue:

A non-zero vector v is an eigenvector of an operator A if there exists a scalar λ such that $Av = \lambda v$. The scalar λ is the eigenvalue for the eigenvector v .

3. (9 points) Complete the following:

a) State the Upper triangular (Schur) representation theorem. Give an example that illustrates this representation.

Let A be the matrix of an operator on an inner product space. Then A can be represented as $A = UTU^*$, where U is unitary and T is upper triangular. In the case of real square matrices, if A has all real eigenvalues then $A = UTU^{-1}$ where U is ^{real} orthogonal and T is real upper triangular.

$$\text{e.g. } \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

b) Give three distinct examples of conditions that are equivalent to the following: a matrix A has 0 as an eigenvalue.

eg. A is not invertible

A has linearly independent columns

A has determinat 0.

c) In some combination, clearly state three ways in which pairs of the following terms connected to a square matrix are related: trace, singular values, eigenvalues, characteristic polynomial, and minimal polynomial.

Many examples are possible. For instance:

The trace of a square matrix is the sum of its eigenvalues.

The minimal polynomial of a square matrix divides its characteristic polynomial.

The roots of a square matrix's characteristic polynomial are its eigenvalues.

The singular values of a matrix A are the positive square roots of the eigenvalues of A^*A .

4. (16 points) Prove **two** of the following:

a) If square matrices A and B are similar, then they have the same rank.

b) For any given real $n \times n$ matrix A and any $\delta > 0$, there exists an ε such that $0 \leq \varepsilon < \delta$ and so that $A - \varepsilon I$ is invertible.

c) For two invertible $n \times n$ matrices A and B , AB and BA have the same eigenvalues, and with the same algebraic multiplicities.

d) Any 2×2 real orthogonal matrix with determinant -1 is similar to $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

First solution:

a) Since A and B are similar, we have $A = PBP^{-1}$ with P invertible. Then $AP = PB$. P operates on A by column operations; P operates on B by row operations. Since row and column operations preserve rank and the two results are equal (and so have equal ranks) we also have that A and B have equal ranks.

b) If A is invertible, take $\varepsilon = 0$ and we are done. Otherwise, A has 0 as an eigenvalue, perhaps with multiplicity

various.

4. (continued) Second solution:

Various

5. (24 points) Always, sometimes, or never? If it is always or never true, give a proof. If it is sometimes true, give an example where it's true and where it's false.

a) A product of two nilpotent matrices is nilpotent.

Sometimes.

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \text{ is nilpotent}$$

$$\text{and } \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

but $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$
 are each nilpotent
 (since strictly U/L triangular)
 and $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix};$
 this is not nilpotent,
 since it has a non-zero
 eigenvalue
 (1).

b) An invertible real 2×2 matrix is matrix equivalent to $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Always.

Every invertible real 2×2 matrix is row equivalent to the identity, since it has full rank. But two matrices that are row equivalent are also matrix equivalent.

c) For a given real symmetric matrix, each of its eigenvalues is real.

Always. This is the content of the Spectral Theorem in the real case.

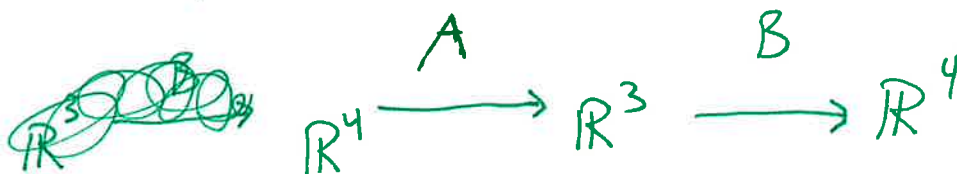
d) If two square matrices each have trace 0, their product also has trace 0.

Sometimes
 $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ has trace 0
 and $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

but
 ~~$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$~~ $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ has trace 0 and
 $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
 has trace 2.

e) If A and B are matrices and A is 3×4 and B is 4×3 , then BA has a left inverse.
 Consider the real case; the same is true over $\mathbb{F}$.

Never

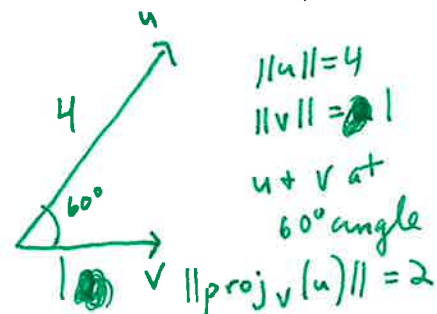


The rank of BA is at most 3, since this is the maximal dimension of the range of A . This means that BA is not injective, and therefore it does not have a left inverse.

f) If the projection of u onto v has norm 2, then the projection of v onto u has norm $1/2$.

Sometimes
 $\|u\|=2$, $\|v\|=1$, $u+v$ colinear.
 $\|proj_v(u)\|=2$

$$\|proj_u(v)\| = 1 \neq \frac{1}{2}$$



$$\|proj_u(v)\| = \frac{1}{2} \checkmark$$

no.

yes.
 or more simply $u = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, $v = \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}$

6. (6 points) Complete the following.

Suppose that for a certain norm on $\mathbb{R}^3$, the vector $\begin{pmatrix} 4 \\ 5 \\ 3 \end{pmatrix}$ has norm 12.

a) Using this information about the norm, compute the norm of $\begin{pmatrix} 8 \\ 10 \\ 6 \end{pmatrix}$ and of $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$; or explain why you do not have enough information to do so.

b) Give an example of a norm on $\mathbb{R}^3$ that has the above property.

c) Explain how your example norm satisfies the defining properties of a norm.

a) norm of $\begin{pmatrix} 8 \\ 10 \\ 6 \end{pmatrix}$ is 24, since norms are linear. (homogeneity)
norm of $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ is 0, by non-degeneracy

b) For instance, the L^1 (taxicab) norm yields that the norm of $\begin{pmatrix} 4 \\ 5 \\ 3 \end{pmatrix}$ is 12

So does the L^3 norm, since $\sqrt[3]{4^3 + 5^3 + 3^3} = 12$

So does the usual Euclidean L^2 norm once it is rescaled by $\frac{12}{5\sqrt{2}}$.

Or the norm $\| \begin{pmatrix} a \\ b \\ c \end{pmatrix} \| = \sqrt{a^2 + 5b^2 + \frac{1}{3}c^2}$

c) various, depending on the norm chosen. properties to verify:

7. (12 points)

a) Circle two matrices that are in Jordan canonical form.

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 1 \\ 0 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} i & 0 & 0 & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & i \end{pmatrix}$$

$$\begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 8 \end{pmatrix}$$

and

b) Circle two matrices that are not diagonalizable over $\mathbb{R}$.

$$\begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 10 & 2 \\ 0 & 10 \end{pmatrix}$$

$$\begin{pmatrix} 5 & 1 \\ 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$$

and

or

total is 66

c) How many real 3×4 matrices are there that are in RREF and so that every entry is either a 1 or a 0?

rank 0: $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

1

rank 1: $\begin{pmatrix} - & - & - & - \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

$2^4 - 1$

two zero rows

any assignment along the first row works (leave out all 0's, since that has already been counted).

rank 2: $\begin{pmatrix} - & - & - & - \\ 0 & - & - & - \\ 0 & 0 & 0 & 0 \end{pmatrix}$

$2^5 - 1 + 2^2$

one zero row.

6 choices for the pair of two pivots

and the boxed number of ways to complete the remaining slots.

$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ 2^4 $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ 2^2

$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ 2^3 $\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ 2^1

$\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ 2^2 $\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ 2^0

$\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$ 2^3 $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$ $10 \cdot 2^2$

$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$ 2^1 $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$ 2^0

rank 3: $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$ $\binom{4}{3} = 4$ choices for pivot columns

$2^4 - 1$

8. (8 points)

Complete **one** of the following.

a) A matrix A_n is an addition table mod n if the element of A in the i th row and j th column a_{ij} is equal to $(i-1) + (j-1) \pmod n$. For example, the addition table mod 4 A_4 looks like this:

$$A_4 = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 0 \\ 2 & 3 & 0 & 1 \\ 3 & 0 & 1 & 2 \end{pmatrix}$$

Show that the determinant of A_n is always equal to 0 mod n .

b) Let the matrix T_n be defined as having entries $a_{ij} = 3^{(i-j)}$. That is, the entry in the i th row and j th column is $3^{(i-j)}$. For all $n \geq 1$, find the determinant and the eigenvalues (with multiplicity) of T_n .

a) Two approaches:

① Compute in mod n . Goal is to show that the determinant of A_n mod n is 0.

For A_4 , notice that we can carry out the following two column operations mod 4, which do not affect the determinant mod 4:

$$A_4 \xrightarrow{\substack{C2 \leftarrow C1 \\ C4 \leftarrow C3}} \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 1 & 3 & 0 \\ 2 & 1 & 0 & 1 \\ 3 & 1 & 1 & 2 \end{pmatrix} \xrightarrow{C4 \leftarrow C3} \begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 1 & 3 & 1 \\ 2 & 1 & 0 & 1 \\ 3 & 1 & 1 & 1 \end{pmatrix}$$

And since the matrix (mod 4) has two equal columns, it has 0 determinant (mod 4).

The same argument works for any pair of columns for any size A_n , since adding one more, when stepping to the right is the definition of the addition (mod n) table.

② Compute only in $\mathbb{Z}$.

By cofactor expansion, it is sufficient to show that A_4 has the same determinant as a matrix where every entry in a single column or row is a multiple of 4.

$$A_4 \xrightarrow{C3 \leftarrow C1} \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 2 & 4 & 0 \\ 2 & 3 & 2 & 1 \\ 3 & 0 & 4 & 2 \end{pmatrix} \xrightarrow{C3 \leftarrow C3 + 2C2} \begin{pmatrix} 0 & 1 & 4 & 3 \\ 1 & 2 & 8 & 0 \\ 2 & 3 & 8 & 1 \\ 3 & 0 & 4 & 2 \end{pmatrix}$$

Since these column moves preserve the determinant, we see that the determinant of A_4 is a multiple of 4.

The argument for A_n proceeds similarly.

9. (9 points) Complete the following.

a) Compute the characteristic polynomial and the minimal polynomial of the following matrix M . Are they equal to each other?

$$M = \begin{pmatrix} 3 & 0 & 1 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

b) There are 9 matrices that can be formed by changing exactly one entry of M to a 2. Give an example of such a matrix that has the same characteristic polynomial as M and the same minimal polynomial as M .

c) Again, there are 9 matrices that can be formed by changing exactly one entry of M to a 2. Give an example of such a matrix that has the same characteristic polynomial as M but a *different* minimal polynomial than M .

a) char poly of M : $(3-\lambda)^3$, since M is upper triangular.
 Since the min poly of M divides the char poly,
 it is either $(3-\lambda)$, $(3-\lambda)^2$, or $(3-\lambda)^3$. M does
 not satisfy $3-\lambda$, but it does satisfy $(3-\lambda)^2$ since
 $(3I-M)^2 = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. So $(3-\lambda)^2$ is the
 minimal poly of M .

b) $\begin{pmatrix} 3 & 0 & 2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

c) $\begin{pmatrix} 3 & 0 & 1 \\ 0 & 3 & 0 \\ 0 & 2 & 3 \end{pmatrix}$, since $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

10. (4 points)

For **one** of the following matrices, compute its 200th power. You should write your answer as a single matrix, but its entries can be left as computations, such as $4 \cdot 5^{100} - 2 \cdot 5^{50}$.

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 6 \end{pmatrix} \quad C = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 1 & 2 & 3 & 5 \\ 0 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

A is a permutation matrix sending
 $e_1 \rightarrow e_3 \rightarrow e_6 \rightarrow e_2 \rightarrow e_5$ and $e_4 \rightarrow e_4$

Since 200 is a multiple of 5, $A^{200} = I_6$.

Since B has 3 distinct eigenvalues (each w/ alg and thus geom mult 1), B is diagonalizable.

$\lambda = 2$ has e-vector $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\lambda = -1$ has e-vector $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

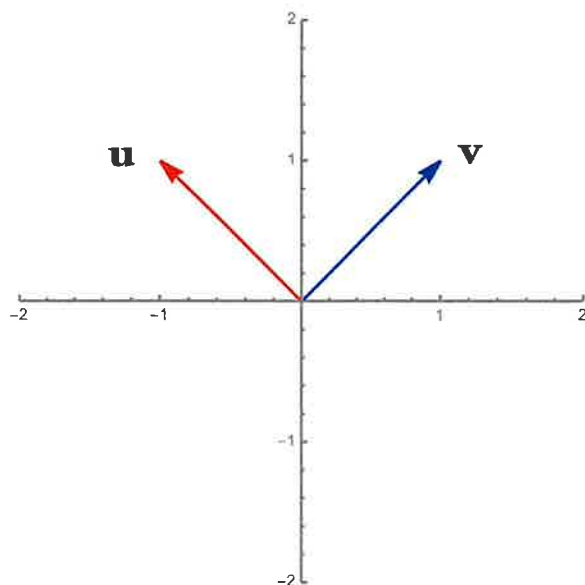
and $\lambda = 6$ has $\text{Nul} \begin{pmatrix} -4 & 0 & 1 \\ 0 & -7 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = z \begin{pmatrix} 1/4 \\ 0 \\ 1 \end{pmatrix}$ or $\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$

(check: $\begin{pmatrix} 2 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 6 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \\ 24 \end{pmatrix}$ ✓)

So $B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 6 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}^{-1}$

And $B^{200} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} 2^{200} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6^{200} \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}^{-1}$

$$= \begin{pmatrix} 2^{200} & 0 & -\frac{2^{200}}{4} + \frac{6^{200}}{4} \\ 0 & 1 & 0 \\ 0 & 0 & 6^{200} \end{pmatrix}$$



11. (6 points)

a) Write the matrix representing the following linear transformation of $\mathbb{R}^2$ using the standard basis:

scale $\mathbf{u}$ by 10 and $\mathbf{v}$ by 8

b) Then compute the singular values of the third power of this matrix.

$$\begin{aligned}
 & a) \quad \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{pmatrix} \begin{pmatrix} 8 & 0 \\ 0 & 10 \end{pmatrix} \begin{pmatrix} \cos 45^\circ & \sin 45^\circ \\ -\sin 45^\circ & \cos 45^\circ \end{pmatrix} \\
 & \text{or} \quad \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 8 & 0 \\ 0 & 10 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = \boxed{\begin{pmatrix} 9 & -1 \\ -1 & 9 \end{pmatrix}}
 \end{aligned}$$

b) since we know the SVD of the matrix from above, we have that the two singular values are 8 and 10. So the singular values of A^3 are 8^3 and 10^3 .

12. (20 points)

Complete the following. Use the next three blank pages, as needed. Label each part of your solution clearly, (a) through (j). (These need not be completed in order.)

Let A be the real matrix $\begin{pmatrix} 2 & 3 & 0 & 0 \\ 3 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

a) Compute the rank and determinant of A . rank = 2, det = 0

b) Compute the characteristic polynomial and eigenvalues of A .

c) Compute bases for the eigenspaces of A .

d) Give the general solution to the matrix equation $Ax = \begin{pmatrix} 4 \\ 1 \\ 0 \\ 0 \end{pmatrix}$.

e) Compute the result of applying A^{99} to the vector $\begin{pmatrix} -5 \\ 5 \\ 0 \\ 0 \end{pmatrix}$.

f) Compute the singular values of A , or explain why this is not possible.

g) Find a real matrix B such that $B^2 = A$, or explain why no such B exists.

h) Give the Jordan canonical form of A ; or explain why A cannot be put in this form.

i) Compute an orthonormal basis for the orthogonal complement of the column space of A . Then compute a different orthonormal basis for the orthogonal complement of the column space of A .

j) Circle the terms that apply to A , and do not circle the ones that don't:

nilpotent invertible unitary triangular diagonalizable permutation symmetric

Solutions for problems 9 & 10:

a) $\text{rank} = 2 \quad \det = 0$

b)
$$\begin{pmatrix} 2-\lambda & 3 & 0 & 0 \\ 3 & 2-\lambda & 0 & 0 \\ 0 & 0 & -\lambda & 0 \\ 0 & 0 & 0 & -\lambda \end{pmatrix} \rightsquigarrow (2-\lambda) \begin{vmatrix} 2-\lambda & 0 & 0 \\ 0 & -\lambda & 0 \\ 0 & 0 & -\lambda \end{vmatrix} - 3 \begin{vmatrix} 3 & 0 & 0 \\ 0 & -\lambda & 0 \\ 0 & 0 & -\lambda \end{vmatrix}$$

$$= (2-\lambda)(2-\lambda)\lambda^2 - 3^2(\lambda^2)$$

$$\lambda^2(\lambda^2 - 4\lambda + 4 - 9) = \lambda^2(\lambda^2 - 4\lambda - 5)$$

$$\lambda^2(\lambda - 5)(\lambda + 1)$$

← char poly.

So $\lambda = 0, 0, 5, -1$.

c) $\lambda = 0: \begin{pmatrix} 2 & 3 & 0 & 0 \\ 3 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$

$\lambda = -1: \begin{pmatrix} 3 & 3 & 0 & 0 \\ 3 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \left\{ \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix} \right\}$

$\lambda = 5: \begin{pmatrix} -3 & 3 & 0 & 0 \\ 3 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\}$

d)
$$\left(\begin{array}{cccc|c} 2 & 3 & 0 & 0 & 4 \\ 3 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \Rightarrow \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ 0 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + w \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Solutions for problems 9 & 10:

e) Since $\begin{pmatrix} -5 \\ 5 \\ 0 \\ 0 \end{pmatrix}$ is in the -1 -eigenspace,

$$\text{we have } A^{99} \begin{pmatrix} -5 \\ 5 \\ 0 \\ 0 \end{pmatrix} = (-1)^{99} \begin{pmatrix} -5 \\ 5 \\ 0 \\ 0 \end{pmatrix} = - \begin{pmatrix} -5 \\ 5 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 5 \\ -5 \\ 0 \\ 0 \end{pmatrix}$$

f) Since A is real, square, and symmetric, its singular values equal its eigenvalues, after taking absolute values. So: $0, 0, 5, 1$

g) not graded.

h) Since $AM=GM$ for all eigenvalues of A , A is diagonalizable. Therefore its JCF is

$$\begin{pmatrix} 5 & & 0 \\ & -1 & \\ 0 & & 0 \\ & & & 0 \end{pmatrix}$$

i) ^{eg.} $\left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$ and $\left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix} \right\}$

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Solutions for problems 9 & 10:

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Solutions for problems 9 & 10:

Bonus: Explain one or more linear algebra ideas or results that you learned in the course but that haven't been asked about on this exam.

various